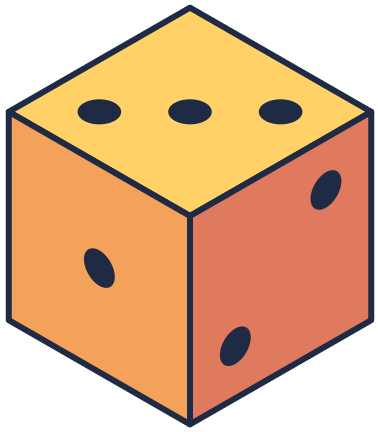
Project 2 | TinkerCad | About 45 min

Make Your Own Dice in TinkerCad

Design a real six-sided die with dots, then 3D print it and play a game with it!



Here's the plan

A die is just a **cube with little holes** in it. In TinkerCad you make holes by putting "hole" shapes into a solid shape and then **grouping** them. We'll do **one side at a time**, always working on the **top** of the die, then flip it over and do the next side. Easy!

You need
Your Chromebook and your TinkerCad class login

Sizes we'll use
Die: `20 × 20 × 20` mm
Dots: `4 × 4 × 4` mm

Words to know

Hole
A see-through shape. When you group it with a solid shape, it cuts that space away.

Group
Joins shapes into one. This is the moment the holes actually get cut.

Align
Lines shapes up perfectly. We use it to put a dot in the exact center.

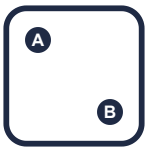
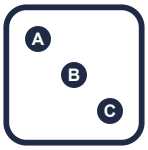
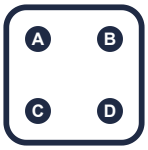
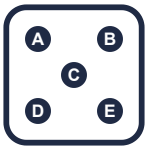
NAME YOUR DESIGN!
Your teacher downloads everyone's dice from TinkerCad to print them. Name yours **Dice - First Name Last Initial** (like **Dice - Maya R**) so we know it's yours. You'll do this in Step 1.

GOLDEN RULE
Made a mistake? Press `Ctrl` + `Z` to undo. You can't break anything. 😄

The Dot Map

The secret rule of dice: opposite sides always add up to **7**. So 1 is across from 6, 2 is across from 5, and 3 is across from 4.

Every dot starts in the **center** of the side. Then you move it with the **arrow keys**. Each press moves it 1 mm, so "5 →" means **press the right arrow 5 times**.

	**1**	across from **6**	**A** stay in center	
	**2**	across from **5**	**A** 5 ← + 5 ↑ **B** 5 → + 5 ↓	
	**3**	across from **4**	**A** 5 ← + 5 ↑ **B** stay in center	**C** 5 → + 5 ↓
	**4**	across from **3**	**A** 5 ← + 5 ↑ **B** 5 → + 5 ↑	**C** 5 ← + 5 ↓ **D** 5 → + 5 ↓
	**5**	across from **2**	**A** 5 ← + 5 ↑ **B** 5 → + 5 ↑ **C** stay in center	**D** 5 ← + 5 ↓ **E** 5 → + 5 ↓
	**6**	across from **1**	**A** 5 ← + 5 ↑ **B** 5 ← **C** 5 ← + 5 ↓	**D** 5 → + 5 ↑ **E** 5 → **F** 5 → + 5 ↓

KEEP THIS PAGE NEXT TO YOU

You'll look back at it for every side. The letters (A, B, C…) are just so you can keep track of which dot you're on.

1 Get set up

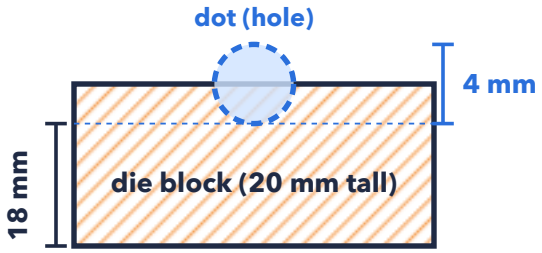
- [ ] Open TinkerCad and click **Create** → **3D Design**.
- [ ] **Name it right away!** Click the silly random name TinkerCad gave your design (top left), type **Dice – First Name Last Initial** (like **Dice – Maya R**), and press `Enter`.
- [ ] In the bottom right corner, find **Snap Grid**. Make sure it says `1.0 mm`.

2 Make the die block

- [ ] Drag a red **Box** from Basic Shapes onto the workplane. It's already `20 × 20 × 20` mm. 👍
- [ ] **Optional, but it looks cool:** in the Shape panel, slide **Radius** up to about `2` to round the corners like real dice.

3 Make your first dot

- [ ] Drag a **Sphere** onto the workplane, next to the die.
- [ ] In the Shape panel, click **Hole** (◌). The sphere turns see-through.
- [ ] Make it smaller: hover over a white corner square, click each number that pops up and type `4`. You want `4 × 4 × 4`.
- [ ] Lift it up: grab the little **black cone** on top of the dot, click the height number and type `18`, then press `Enter`.



Why 18? The die is 20 mm tall. Lifting the 4 mm dot to 18 puts it **halfway into** the top, so it makes a nice round dimple.

Move it to the exact center

- [ ] Drag a box around the die **and** the dot so both are selected.
- [ ] Click **Align** (⊞) (or press `L`). Small dots appear around your shapes.
- [ ] Click the **middle** dot on the line along the **front**, and the **middle** dot on the line along the **side**. **Don't** click the up-and-down line, or your dot will lose its height!

CHECK
Your dot should sit right in the middle of the top, half sunk in. That's side 1! 🎉

4 Add the other dots

Side 1 only needs one dot, so you can skip this step the first time. For every other side, start with a fresh center dot (Step 3), then:

- [ ] Click **Top** on the view cube (the little cube in the top left) so you're looking straight down. Now the arrow keys go the same way as the arrows on the Dot Map.
- [ ] Click the center dot and press `Ctrl` + `D` (Duplicate). A copy appears right on top of it.
- [ ] Use the **arrow keys** to move the copy. Check the **Dot Map** for how many presses. Example for dot A on side 4: press `←` 5 times and `↑` 5 times.
- [ ] Repeat until you have every dot for that side.
- [ ] If this side has **no** middle dot (sides 2, 4 and 6), click the center dot and press `Backspace`.

TIP
Click the view cube's **Home** button (the little house) any time to see the die from the side again. Look closely: all the dots should be at the **same height**.

5 Group to cut the holes

- [ ] Press `Ctrl` + `A` to select everything (the die and this side's dots).
- [ ] Click **Group** (or press `Ctrl` + `G`).

CHECK
The see-through spheres disappear and you have real dimples in the top of your die. 🙌 The die is now one shape, so it's safe to flip it.

UH OH?
If a dot went to the wrong place, press `Ctrl` + `Z` to undo the group, fix the dot, then group again.

6 Flip it and do the next side

We always add dots to the **top**. So after each side, we turn the die until a **blank** side is on top. Do the sides in **this order** and the opposite sides will always add up to 7 automatically!

How to turn the die

- ☐ Click the die. You'll see three **curved arrows**. Two are on the **sides** (those flip it over). The one near the bottom just spins it in place, so don't use that one.
- ☐ Drag a side arrow around until the number says the angle in the chart below. You can also click the number and type it.
- ☐ If the die ends up floating or sunk into the workplane, press `D` to drop it back down.

1



Side 1

–

Start here. Nothing to rotate yet.

2



Side 6

180°

Pick one curved arrow on the **side** of the die. Turn it **180°**.

3



Side 2

90°

Same arrow. Turn it **90°**.

4



Side 5

180°

Same arrow. Turn it **180°**.

5



Side 3

90°

Now use the **other** side arrow. Turn it **90°**.

6



Side 4

180°

Same arrow as before (the other one). Turn it **180°**.

CHECK AFTER EVERY TURN

The top should be **completely blank**. If you see dots on top, undo (`Ctrl` + `Z`) and try the other side arrow.

Then for each side: **Turn ➜ Make a center dot** (Step 3) **➜ Add dots** (Step 4) **➜ Group** (Step 5) **➜** repeat!

7 Final check ✅

Click the view cube's **Home** button, then drag the view cube to spin around and look at every side. Check each box:

- ☐ All six sides have dots: 1, 2, 3, 4, 5 and 6.
- ☐ Opposite sides add up to 7 (1 & 6, 2 & 5, 3 & 4).
- ☐ All the dots are dimples (holes), not bumps sticking out.
- ☐ It's still about `20 × 20 × 20` mm and sitting flat on the workplane.

8 Save it and hand it in 💾

- ☐ Check the name at the top left says **Dice – First Name Last Initial**. Not right? Click it and fix it.
- ☐ TinkerCad saves by itself while you work. To be sure, click the **TinkerCad logo** (top left) to go back to your designs. Your die should be there with **your name on it**.
- ☐ Tell your teacher your die is ready! Your teacher will download it from TinkerCad and 3D print it for you. 🖨️
- ☐ **Done early?** Make it yours! Try a different color, rounder corners, or a tiny shape on side 1 instead of a dot (like a star or your initial). Keep the dots at least `3` mm wide so they show up when printed.

🤔 Think about it

Real designers ask themselves these questions after every project.

1. What was the hardest part, and how did you figure it out?

2. If you made another die, what would you do differently?

CREDITS

This guide is adapted from "How to Make Your Own 6-sided Dice on TinkerCad" by **Potato_Person_** on Instructables, used under a Creative Commons BY-NC-SA license. Thank you, Potato_Person_! **Changes:** rewritten as a step-by-step classroom worksheet, with new drawings, a dot map, exact sizes (20 mm die, 4 mm dots) and a "work on the top, then flip" method. This adapted version is shared under the same license, CC BY-NC-SA 4.0: free to share and change for non-commercial use, as long as you give credit and share alike. TinkerCad is a trademark of Autodesk; this guide is not made by or affiliated with Autodesk or Instructables.